

# Master Telemetry Matrix & Engineering Tolerances

Document ID: SPEC-TEL-07 | Systems Engineering Reference Table Matrix

## 1. Atmospheric Sensors & Life Support Parameters

| Subsystem Sensor       | Nominal | Min Alarm | Max Alarm | Unit |
|------------------------|---------|-----------|-----------|------|
| Habitat Pressure       | 101.3   | 95.0      | 105.0     | kPa  |
| Oxygen Partial (pO2)   | 21.0    | 19.5      | 23.5      | kPa  |
| Carbon Dioxide (pCO2)  | 0.35    | 0.0       | 0.70      | kPa  |
| Nitrogen Partial (pN2) | 78.0    | 74.0      | 81.0      | kPa  |
| Relative Humidity      | 45.0    | 30.0      | 60.0      | %    |
| Cabin Ambient Temp     | 21.5    | 18.0      | 25.0      | degC |

## 2. Cryogenic Propellant Storage Telemetry

| Tank Identifier | Cryogen         | Quantity (kg) | Storage Temp (K) | Boil-off Rate |
|-----------------|-----------------|---------------|------------------|---------------|
| TK-LOX-01       | Liquid Oxygen   | 18500         | 90.2             | 0.05% / sol   |
| TK-LOX-02       | Liquid Oxygen   | 18500         | 90.1             | 0.04% / sol   |
| TK-CH4-01       | Liquid Methane  | 6200          | 111.5            | 0.08% / sol   |
| TK-CH4-02       | Liquid Methane  | 6200          | 111.4            | 0.07% / sol   |
| TK-LN2-01       | Liquid Nitrogen | 4500          | 77.3             | 0.02% / sol   |

## 3. Nuclear & Solar Power Bus Voltage & Amperage

| Power Bus ID | Source Type         | Voltage (VDC) | Max Current (A) | Efficiency |
|--------------|---------------------|---------------|-----------------|------------|
| BUS-NUC-A    | Kilopower Stirling  | 120.0         | 83.3            | 94.2%      |
| BUS-NUC-B    | Kilopower Stirling  | 120.0         | 83.3            | 94.0%      |
| BUS-SOL-01   | Photovoltaic Wing 1 | 240.0         | 41.6            | 29.5%      |
| BUS-SOL-02   | Photovoltaic Wing 2 | 240.0         | 41.6            | 29.8%      |
| BUS-BAT-PRI  | Lithium-Sulfur Pack | 120.0         | 150.0           | 98.1%      |